Slick Trace investigation report — case_0009

Generated 2026-09-25T18:34:06.914058+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

Leading candidate: 419630599. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

Scene

Field	Value
Incident	case_0009
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
unassessed	2
candidate	4

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.9125 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	e0ace37901d761a8
coverage_result.json	ac44aef9fa9bf194
forcing.npz	75abcfbcc618a9f4
forecast_result.json	c1b360eaba787cf3
fragility_result.json	f7623e681b81773a
historical_mapping.json	c0ca27295dcca6f6
manifest.json	edc1af030f4b944b
mask_groundtruth.png	877154e84e621d61
naive_baseline_result.json	6d56a8ff2675affd
next_observation_result.json	851e0a03826ad1f7
scene.png	e8498989f80fa09b

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.